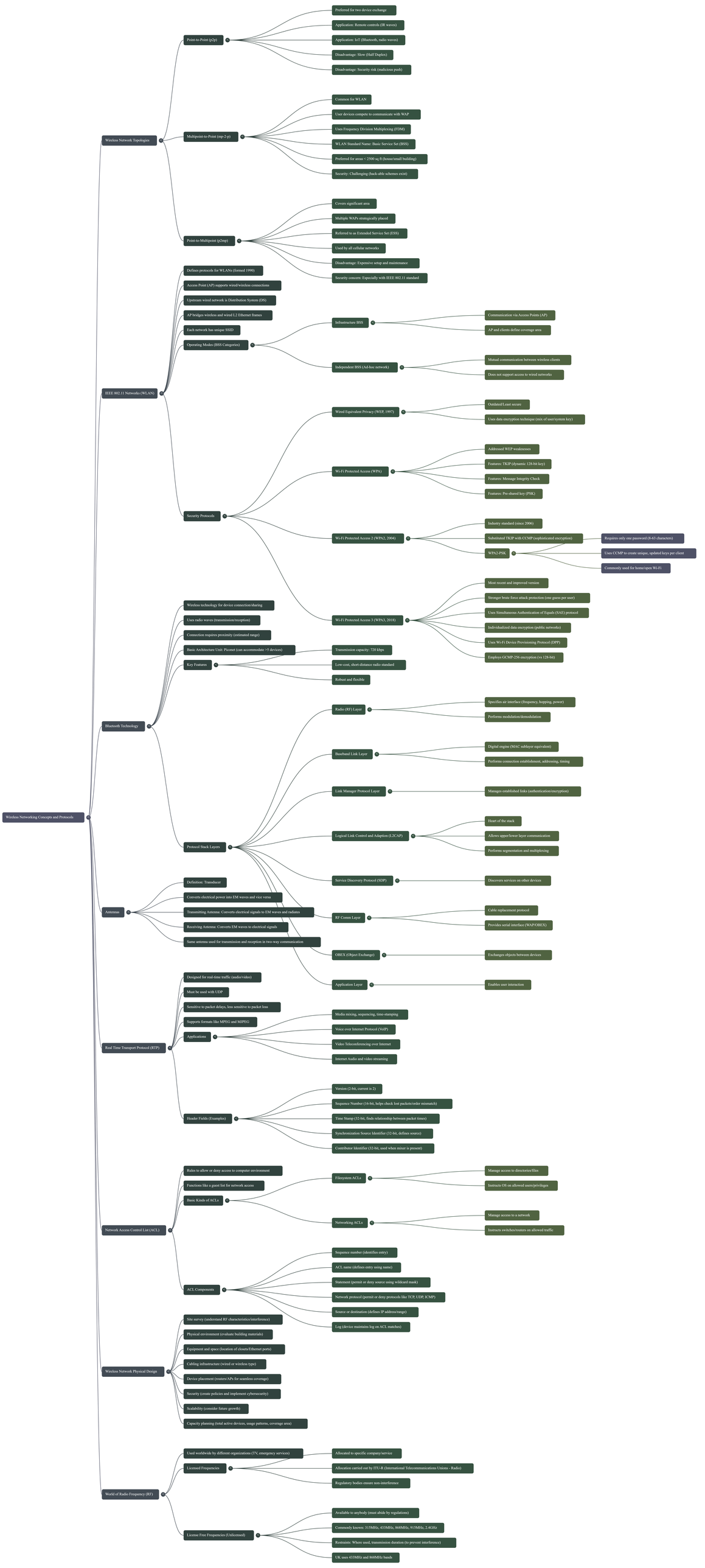
Wireless Networking Concepts and Protocols
Wireless Network Topologies
Point-to-Point (p2p)
Preferred for two device exchange
Application: Remote controls (IR waves)
Application: IoT (Bluetooth, radio waves)
Disadvantage: Slow (Half Duplex)
Disadvantage: Security risk (malicious push)
Multipoint-to-Point (mp-2-p)
Common for WLAN
User devices compete to communicate with WAP
Uses Frequency Division Multiplexing (FDM)
WLAN Standard Name: Basic Service Set (BSS)
Preferred for areas < 2500 sq ft (house/small building)
Security: Challenging (hack-able schemes exist)
Point-to-Multipoint (p2mp)
Covers significant area
Multiple WAPs strategically placed
Referred to as Extended Service Set (ESS)
Used by all cellular networks
Disadvantage: Expensive setup and maintenance
Security concern: Especially with IEEE 802.11 standard
IEEE 802.11 Networks (WLAN)
Defines protocols for WLANs (formed 1990)
Access Point (AP) supports wired/wireless connections
Upstream wired network is Distribution System (DS)
AP bridges wireless and wired L2 Ethernet frames
Each network has unique SSID
Operating Modes (BSS Categories)
Infrastructure BSS
Communication via Access Points (AP)
AP and clients define coverage area
Independent BSS (Ad-hoc network)
Mutual communication between wireless clients
Does not support access to wired networks
Security Protocols
Wired Equivalent Privacy (WEP, 1997)
Outdated/Least secure
Uses data encryption technique (mix of user/system key)
Wi-Fi Protected Access (WPA)
Addressed WEP weaknesses
Features: TKIP (dynamic 128-bit key)
Features: Message Integrity Check
Features: Pre-shared key (PSK)
Wi-Fi Protected Access 2 (WPA2, 2004)
Industry standard (since 2006)
Substituted TKIP with CCMP (sophisticated encryption)
WPA2-PSK
Requires only one password (8-63 characters)
Uses CCMP to create unique, updated keys per client
Commonly used for home/open Wi-Fi
Wi-Fi Protected Access 3 (WPA3, 2018)
Most recent and improved version
Stronger brute force attack protection (one guess per user)
Uses Simultaneous Authentication of Equals (SAE) protocol
Individualized data encryption (public networks)
Uses Wi-Fi Device Provisioning Protocol (DPP)
Employs GCMP-256 encryption (vs 128-bit)
Bluetooth Technology
Wireless technology for device connection/sharing
Uses radio waves (transmission/reception)
Connection requires proximity (estimated range)
Basic Architecture Unit: Piconet (can accommodate >5 devices)
Key Features
Transmission capacity: 720 kbps
Low-cost, short-distance radio standard
Robust and flexible
Protocol Stack Layers
Radio (RF) Layer
Specifies air interface (frequency, hopping, power)
Performs modulation/demodulation
Baseband Link Layer
Digital engine (MAC sublayer equivalent)
Performs connection establishment, addressing, timing
Link Manager Protocol Layer
Manages established links (authentication/encryption)
Logical Link Control and Adaption (L2CAP)
Heart of the stack
Allows upper/lower layer communication
Performs segmentation and multiplexing
Service Discovery Protocol (SDP)
Discovers services on other devices
RF Comm Layer
Cable replacement protocol
Provides serial interface (WAP/OBEX)
OBEX (Object Exchange)
Exchanges objects between devices
Application Layer
Enables user interaction
Antennas
Definition: Transducer
Converts electrical power into EM waves and vice versa
Transmitting Antenna: Converts electrical signals to EM waves and radiates
Receiving Antenna: Converts EM waves to electrical signals
Same antenna used for transmission and reception in two-way communication
Real Time Transport Protocol (RTP)
Designed for real-time traffic (audio/video)
Must be used with UDP
Sensitive to packet delays, less sensitive to packet loss
Supports formats like MPEG and MJPEG
Applications
Media mixing, sequencing, time-stamping
Voice over Internet Protocol (VoIP)
Video Teleconferencing over Internet
Internet Audio and video streaming
Header Fields (Examples)
Version (2-bit, current is 2)
Sequence Number (16-bit, helps check lost packets/order mismatch)
Time Stamp (32-bit, finds relationship between packet times)
Synchronization Source Identifier (32-bit, defines source)
Contributor Identifier (32-bit, used when mixer is present)
Network Access Control List (ACL)
Rules to allow or deny access to computer environment
Functions like a guest list for network access
Basic Kinds of ACLs
Filesystem ACLs
Manage access to directories/files
Instructs OS on allowed users/privileges
Networking ACLs
Manage access to a network
Instructs switches/routers on allowed traffic
ACL Components
Sequence number (identifies entry)
ACL name (defines entry using name)
Statement (permit or deny source using wildcard mask)
Network protocol (permit or deny protocols like TCP, UDP, ICMP)
Source or destination (defines IP address/range)
Log (device maintains log on ACL matches)
Wireless Network Physical Design
Site survey (understand RF characteristics/interference)
Physical environment (evaluate building materials)
Equipment and space (location of closets/Ethernet ports)
Cabling infrastructure (wired or wireless type)
Device placement (routers/APs for seamless coverage)
Security (create policies and implement cybersecurity)
Scalability (consider future growth)
Capacity planning (total active devices, usage patterns, coverage area)
World of Radio Frequency (RF)
Used worldwide by different organizations (TV, emergency services)
Licensed Frequencies
Allocated to specific company/service
Allocation carried out by ITU-R (International Telecommunications Unions - Radio)
Regulatory bodies ensure non-interference
License Free Frequencies (Unlicensed)
Available to anybody (must abide by regulations)
Commonly known: 315MHz, 433MHz, 868MHz, 915MHz, 2.4GHz
Restraints: Where used, transmission duration (to prevent interference)
UK: uses 433MHz and 868MHz bands